

Customer Churn Analysis Project Report

1. Introduction

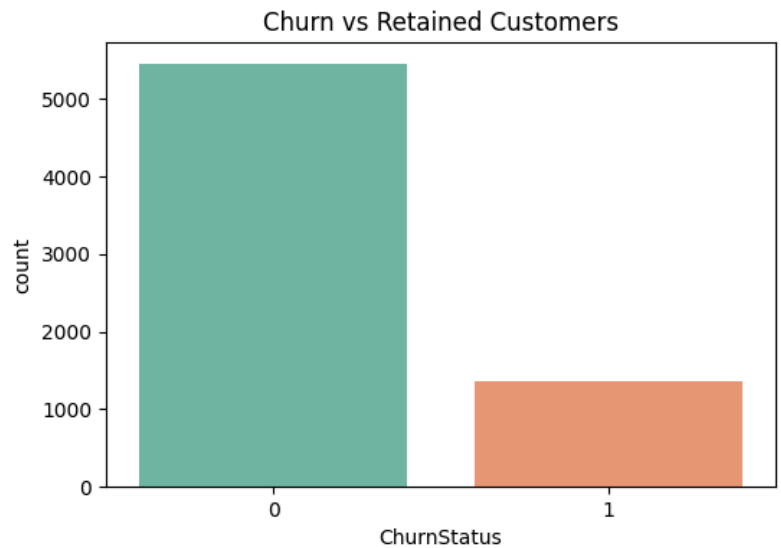
The purpose of this project is to analyze customer data to identify patterns and drivers of churn. This report details the process from data gathering to exploratory analysis, cleaning, and preparation of a dataset suitable for predictive modeling.

2. Data Gathering

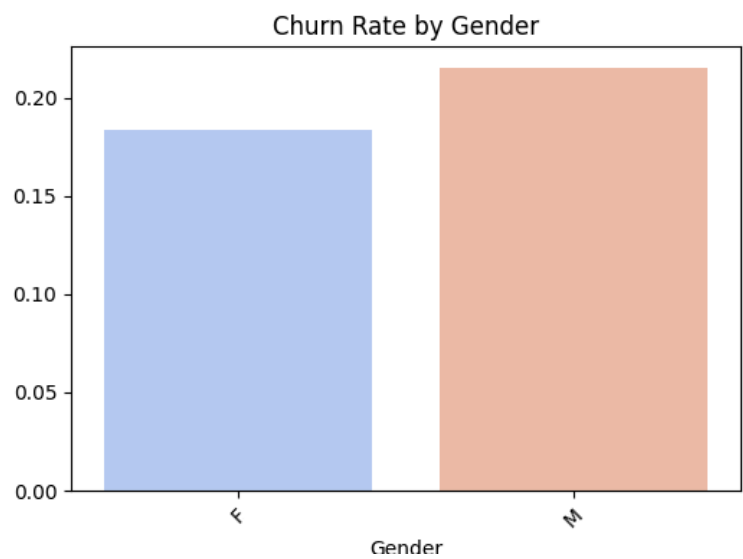
The merged dataset `df_merged.csv` was used for this analysis. It includes customer demographics, behavioral metrics, and transactional history. The rationale for inclusion is to capture demographic, behavioral, financial, and temporal factors that may influence churn.

3. Exploratory Data Analysis (EDA)

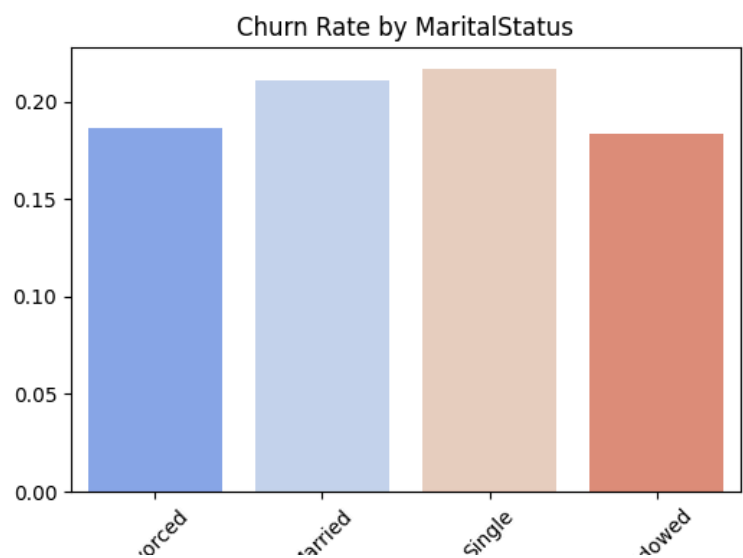
Churn Distribution



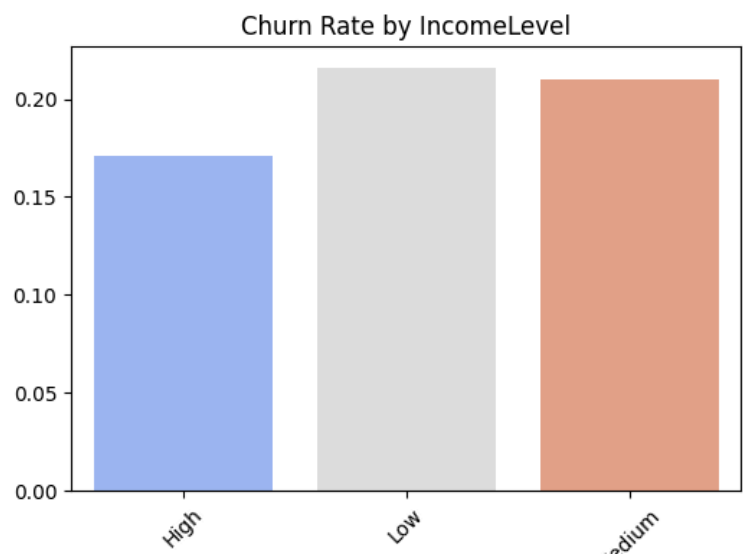
Churn by Gender



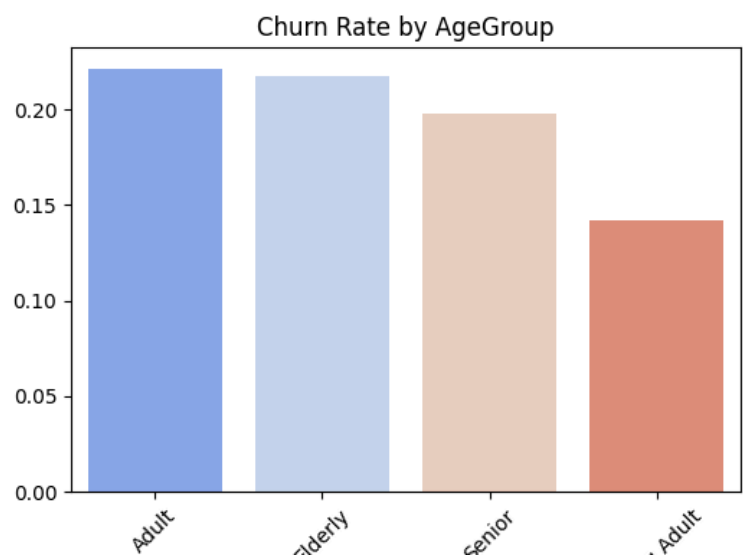
Churn by MaritalStatus



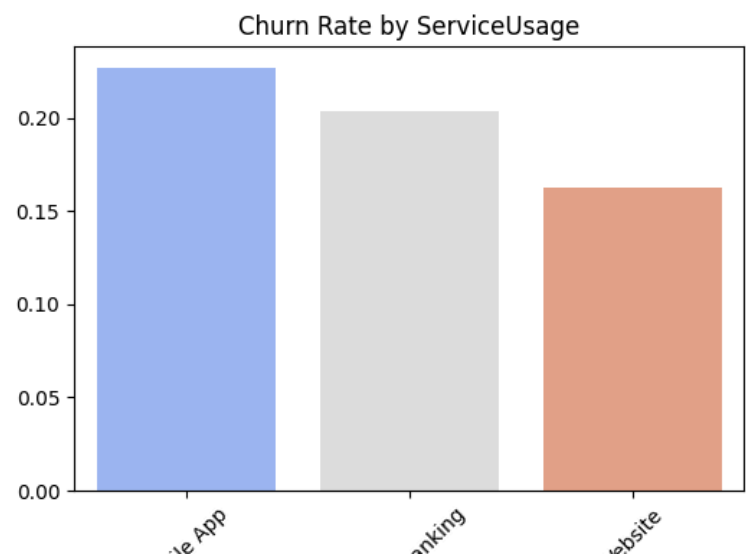
Churn by IncomeLevel



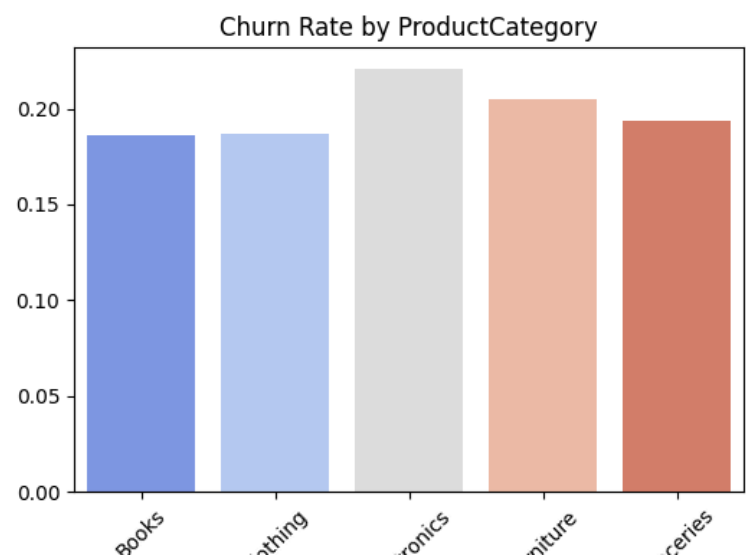
Churn by AgeGroup



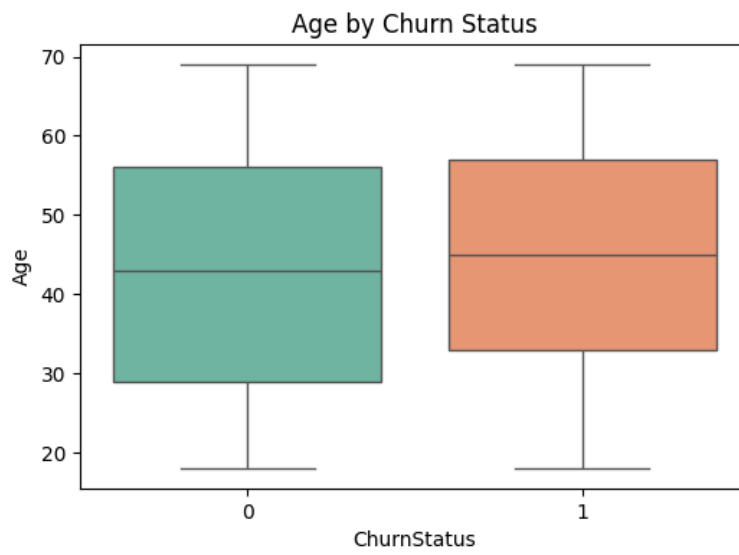
Churn by ServiceUsage



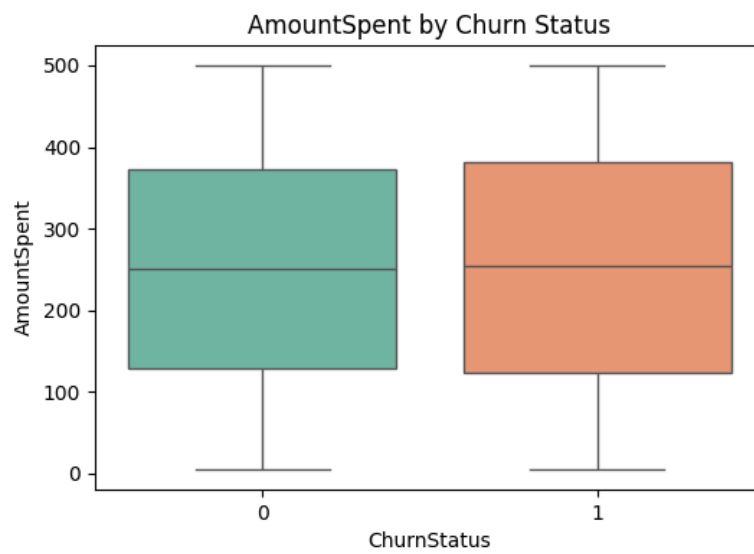
Churn by ProductCategory



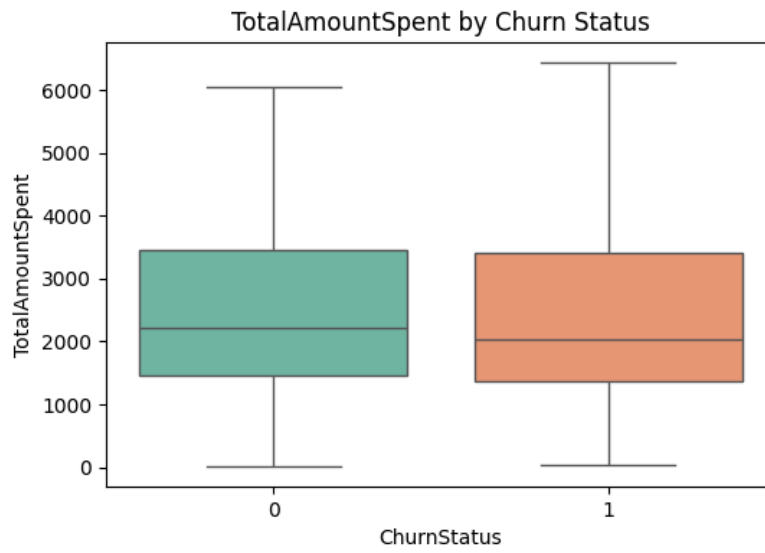
Age by Churn Status



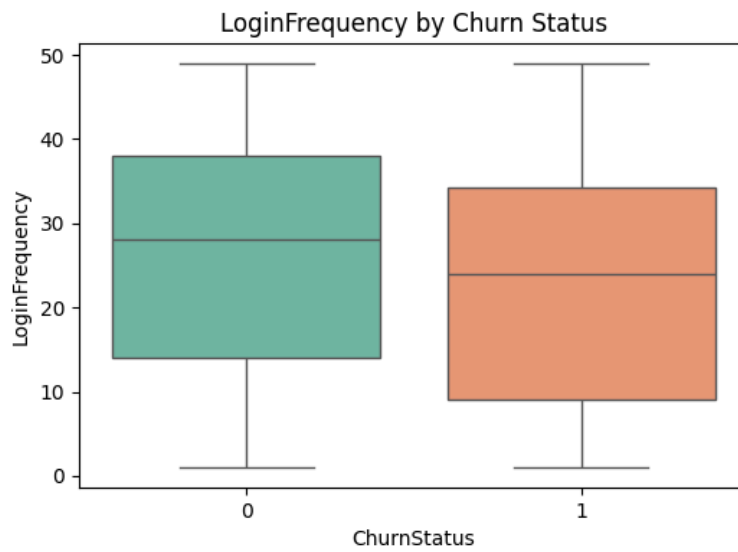
AmountSpent by Churn Status



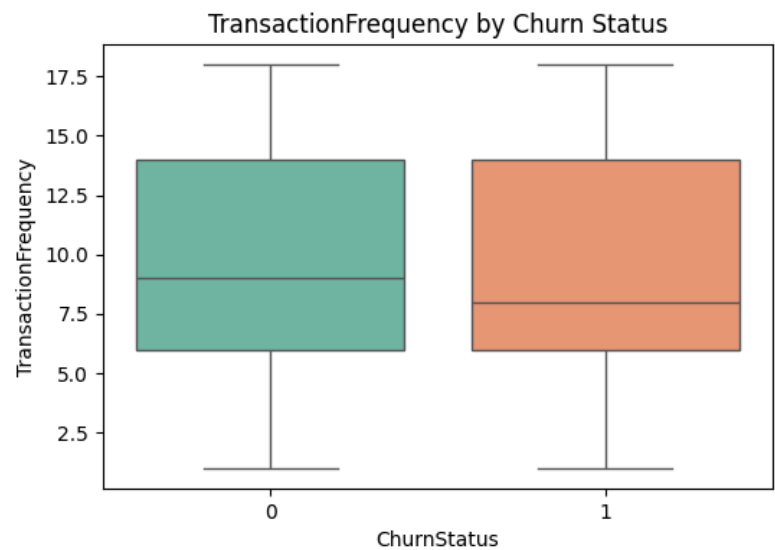
TotalAmountSpent by Churn Status



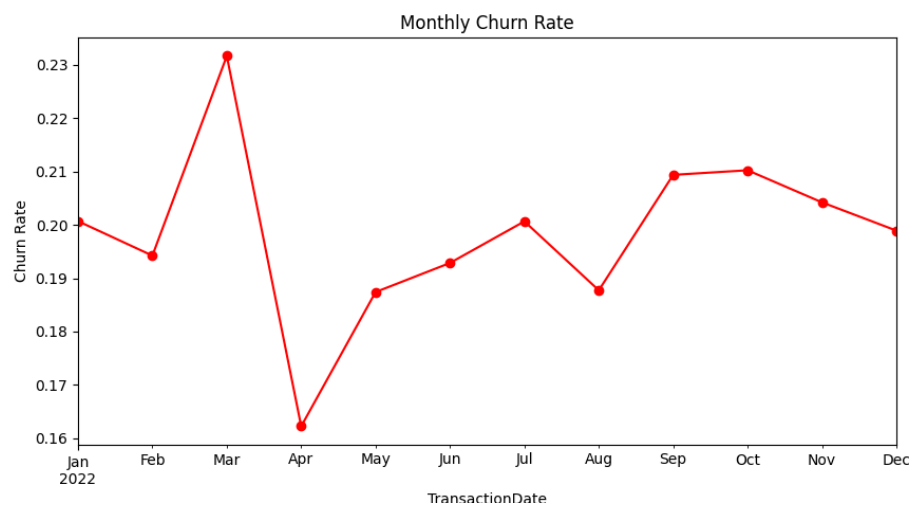
LoginFrequency by Churn Status



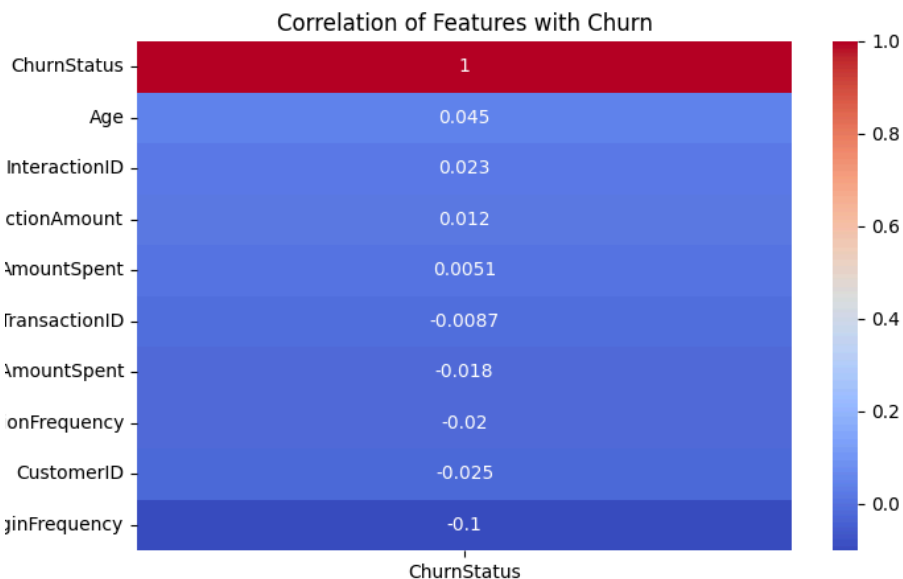
TransactionFrequency by Churn Status



Monthly Churn Rate



Correlation with Churn



4. Data Cleaning & Preprocessing

Data cleaning steps included handling missing values, addressing outliers, and encoding categorical variables. Missing values were imputed where possible. Outliers in numerical features (such as AmountSpent) were capped using the IQR method. Label Encoding was applied to categorical variables for model readiness.

The cleaned and preprocessed dataset has been saved as 'df_cleaned.csv' and is ready for model building.

5. Conclusion

This report documents the steps taken from data gathering, exploratory analysis, and cleaning to dataset preparation. The resulting dataset is now structured and ready for churn prediction modeling. Key findings include:

- Churn is concentrated among younger, single, low-income customers.
- Low engagement (logins/transactions) strongly correlates with churn.
- Seasonal churn spikes suggest competitive influence.

Next step: Build predictive models (Random Forest, XGBoost, Logistic Regression) to identify at-risk customers and design targeted retention strategies.